Sean L. Snaider

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EDUCATION

Northeastern University — **Boston, Massachusetts**

B.S. in Computer Science, Concentration in Software Engineering — *Expected May 2028*

- **GPA:** 3.89/4.00, *Dean's List*

SKILLS & INTERESTS

Languages: Python, Java, TypeScript, JavaScript, C, SQL, Rust
AI / LLM: Anthropic API (structured outputs, tool use), Claude Code, GitHub Copilot, agentic coding workflows
Frameworks & Data: React, Next.js, Tailwind CSS, FastAPI, Prisma, PostgreSQL, MongoDB, Pandas, NumPy, Polars
Tools & Infrastructure: Git, Docker, CMake, GitHub Actions, Linux/WSL, Vite, Vercel, GDB
Interests: Speedcubing (10+ years, 6.14s single), Guitar (progressive and math rock)

EXPERIENCE

Vertex Pharmaceuticals — **Boston, Massachusetts**

G&A Architecture and Strategy Co-op — Jul 2026 - Dec 2026

- Building a Python tool to automate data migrations previously run by hand across multiple platforms and a legacy Microsoft Access workflow; projected by management to eliminate ~$100K in annual effort across business and IT teams.
- Designed the tool so semi-technical staff can reconfigure migrations themselves without code changes, driving field mappings from declarative Excel configuration with per-field overrides rather than hardcoded transformation logic.
- Added checks that migrated records are correct, not just present: value fidelity and referential integrity.

Student Government Association, Northeastern University — **Boston, Massachusetts**

Software Engineer — *Feb 2026 - Present*

- Replaced spreadsheet-based attendance and voting workflows with a full-stack web application now used by 50+ SGA members across 20+ officer meetings per semester.
- Hardened vote handling for SGA executive officer elections by moving validation from the browser to the server with Next.js API routes, Prisma, and schema-checked requests, closing a class of data-integrity bugs.

Khoury College of Computer Sciences, Northeastern University — **Boston, Massachusetts**

Teaching Assistant — *Sep 2025 - Apr 2026*

- Supported 850+ students across two of Khoury's largest intro CS courses through weekly debugging hours covering OOP, data structures, recursion, and control flow.
- Coached students on effective use of Claude Code and GitHub Copilot in lab sessions and one-on-one, covering how to scope tasks, supply relevant context, and iterate when generated output missed intent.

PROJECTS

BellyDSP: Open-Source Guitar Amp Simulator - *C++20, JUCE, Python, CMake, GitHub Actions* - bellydsp.com — *Sep 2026 - Present*

- Built and shipped a free guitar amp app for Mac and Windows: designed it, hand-built a Rust prototype, then directed parallel Claude Code agents to rebuild it in C++20, backed by ~7,200 automated tests.
- Enforced real-time reliability with tests proving the live audio path never allocates memory or waits on locks, the most common causes of audio dropouts.
- Made the app run on weaker computers by cutting CPU use about 4x, running only the selected amp and making its neural network code 2x faster, while keeping amp switches seamless.
- Built a feature that recreates a song's guitar tone with the app's own gear, separating the guitar from the mix with the Demucs AI model; a 50-case benchmark decided which improvements shipped, cutting matching error 31% (statistically significant, $p < 0.0001$).

Rubik's Cube Solver - *Python, React, TypeScript, FastAPI, MongoDB - Github* — *Feb 2026*

- Built a solver for any scramble (≤20 moves, under 1s), implementing Kociemba's two-phase algorithm from the papers.
- Traced intermittent failed solves to a fixed-orientation assumption by testing scrambles on a physical cube, then normalized every state to white top, green front before solving.
- Shipped it as a web app with a CSS 3D cube and a WCA-style timer, now used by 10+ peers for timed solves.

Systems Programming (Coursework) - C, x86-64 Assembly, Linux — *Fall 2025*

- Built operating-system building blocks in C: a FUSE filesystem driver over a disk image with bitmap block allocation, a first-fit free-list memory allocator, and a Unix shell with pipes and I/O redirection.